Round 2, item 01: the linear algebra of genus-3 IC-2LP after the cubic GHS transfer

2026-10-15. Round rules: ../RULES.md. Protocol, frozen before data, with amendments A1-A5: PROTOCOL.md. Prior work: item 04. Tool: `examples/ic_improve2_01_genus3_linalg.rs` (a copy of `examples/ic_improve_04_genus3_2lp.rs`; `src/` is not touched). Analysis: `examples/ic_improve2_01_analyze.rs`. Figure: plot.py, which only plots `summary.json`. Records: results/run/; full tables: results/run/analysis.md.

Verdict

The incumbent is item 04's IC-2LP: kappa = 2, large primes removed by DFS spanning-tree substitution, then Lanczos on the normal equations (`main:inc`). The candidate frozen by tuning is **`lpsgewd`**: the large primes are kept as matrix columns, structured Gaussian elimination (SGE) merges them away, and scalar Wiedemann solves the square remainder. Fits are over q = 2^10..2^20, 3 curves per size, paired on instance, target and walk.

level	result	verdict
exponent, whole solve	q^1.380 [1.376, 1.384] against q^1.398 [1.394, 1.402]; Delta **-0.018 [-0.020, -0.016]**	**ADVANCES** (small)
constant, whole solve	paired ratio at 2^20 **0.653 [0.647, 0.660]**; below 1 at every size (0.76 at 2^10)	**ADVANCES**
primitive (linear algebra alone)	LA ratio at 2^20 **0.463 [0.456, 0.470]**; LA exponent 1.702 against 1.519, Delta **+0.184 [+0.179, +0.189]**	**TRADES**: cheaper at every size, but by less as q grows
sensitivity, `c_ell = 9`	ratio at 2^20 0.494 [0.486, 0.501]; whole exponent 1.563 against 1.491, Delta +0.071 [+0.068, +0.075]	TRADES (constant ADVANCES, exponent REGRESSES)

Does the whole exponent move toward 4/3? Yes, but only a little.

- It passes the protocol's rule:
  - Delta over 10..20 has its CI below 0.
  - The 14..20 slope falls from 1.425 [1.422, 1.428] to **1.415 [1.413, 1.417]**, with a paired difference of -0.011 [-0.013, -0.008].
  - That difference is just outside the A/A noise floor over 14..20, which is about 0.008 (see Controls).
- The local exponent at 19-20 falls from 1.464 [1.446, 1.478] to 1.423 [1.417, 1.429].
- It does **not** settle at 4/3. The trend of the local exponents is still +0.020 per bit [+0.017, +0.025], against +0.018 [+0.014, +0.022] for the incumbent.

Why the linear algebra stays near q^1.5.

- `lpsgewd` holds the solver's row weight flat at about 15 nonzeros per column. The incumbent's weight climbs from 7.3 to 25.3.
- But SGE removes a shrinking share of the columns. The solver keeps 32% of the incumbent's column count at 2^10 and 91% at 2^20.
- So the weight growth is traded for column growth. At 19-20 both arms' linear algebra has the same local exponent, 1.543.
- The LA ratio flattens at 0.45-0.46 over 2^18 to 2^20 and creeps up to 0.479 [0.470, 0.488] at 2^21.
- The linear-algebra exponent over 10..20 is higher only because SGE's advantage is largest at small q.

Exploratory (amendment A3): kappa re-tuned to 1.41. This value came from the training-curve optimum for the incumbent's own linear algebra.

- With the new linear algebra (`tuned:lpsgewd`):
  - **the whole exponent is 1.357 [1.354, 1.360]**, Delta -0.041 [-0.043, -0.039];
  - the 14..20 slope is **1.375 [1.373, 1.378]**, against 1.425;
  - the local exponent at 19-20 is **1.379 [1.376, 1.381]**;
  - the trend is +0.011 per bit [+0.008, +0.013];
  - at `c_ell = 9` the exponent is 1.493, Delta +0.002 [-0.002, +0.005], so it matches.
- Its paired ratio falls monotonically, from 0.87 at 2^10 to 0.659 [0.653, 0.667] at 2^20.
- It is not cheaper than `main:lpsgewd` at any size up to 2^20; the two tie at 2^20.
- At 2^21 it is cheaper (see the 2^21 extension).
- kappa re-tuned alone, with the incumbent's linear algebra (`tuned:inc`):
  - exponent 1.373, Delta -0.025 [-0.028, -0.022];
  - ratio 0.850 [0.840, 0.855] at 2^20;
  - this is the cheapest change to make, and it moves the exponent more than the new linear algebra at kappa = 2 does.
- These are exploratory arms. The protocol's frozen rule (the kappa that minimises `lpsgewd` at b = 20 on training curves) chose kappa = 2.

The other linear-algebra arms (whole cost; paired against `main:inc`):

arm	rows / SGE / solver	whole exponent Delta [95%]	ratio at the top size [95%]	LA ratio at the top size	verdict (exponent; constant)
`wied`	incumbent / no / Wiedemann	+0.008 [+0.007, +0.009]	0.939 [0.938, 0.941] (2^20)	0.906	REGRESSES; ADVANCES (TRADES)
`sgewd`	incumbent / yes / Wiedemann	+0.024 [+0.023, +0.025]	0.896 [0.896, 0.897] (2^20)	0.839	REGRESSES; ADVANCES (TRADES)
`sgelz`	incumbent / yes / Lanczos	+0.027 [+0.026, +0.028] (10-18)	0.904 [0.903, 0.907] (2^18)	0.837	REGRESSES; ADVANCES (TRADES)
`bfslz`	BFS basis / no / Lanczos	**-0.029 [-0.031, -0.026]** (10-18)	0.835 [0.830, 0.842] (2^18)	0.719	ADVANCES; ADVANCES
`bfswd`	BFS basis / no / Wiedemann	-0.024 [-0.026, -0.022] (10-18)	0.767 [0.762, 0.772] (2^18)	0.602	ADVANCES; ADVANCES
`bfssgelz`	BFS basis / yes / Lanczos	-0.002 [-0.004, 0.000] (10-18)	0.760 [0.755, 0.763] (2^18)	0.590	MATCHES; ADVANCES
`bfssgewd`	BFS basis / yes / Wiedemann	-0.010 [-0.012, -0.009]	0.694 [0.689, 0.701] (2^20)	0.527	ADVANCES; ADVANCES
`lpsgelz`	LP columns / yes / Lanczos	-0.009 [-0.011, -0.007]	0.700 [0.692, 0.707] (2^20)	0.535	ADVANCES; ADVANCES

arm	rows / SGE / solver	whole exponent Delta [95%]	ratio at the top size [95%]	LA ratio at the top size	verdict (exponent; constant)
`lpsgewd`	LP columns / yes / Wiedemann	**-0.018 [-0.020, -0.016]**	**0.653 [0.647, 0.660]** (2^20)	0.463	ADVANCES; ADVANCES

- Only one arm lowers the linear-algebra exponent itself: **`bfslz`**.
  - It re-bases the cycle space on a breadth-first spanning tree, so the cycles are shorter.
  - Its LA exponent is 1.475 [1.470, 1.480] over 10..18, Delta -0.045 [-0.053, -0.036].
  - Its row weight at 2^18 is 12.8 against 19.2.
  - The incumbent's DFS substitution makes long cycle rows, and that is the source of the q^0.18 weight growth item 04 measured.
- SGE then buys a large constant at small q that shrinks with q, and that raises every SGE arm's fitted LA exponent.
- Wiedemann alone is only a constant gain, about 6-10% of the linear algebra:
  - Lanczos never formed `A^T A` explicitly;
  - both solvers use about `2 n nnz` products;
  - Wiedemann saves the vector updates and the excess rows.

Correctness.

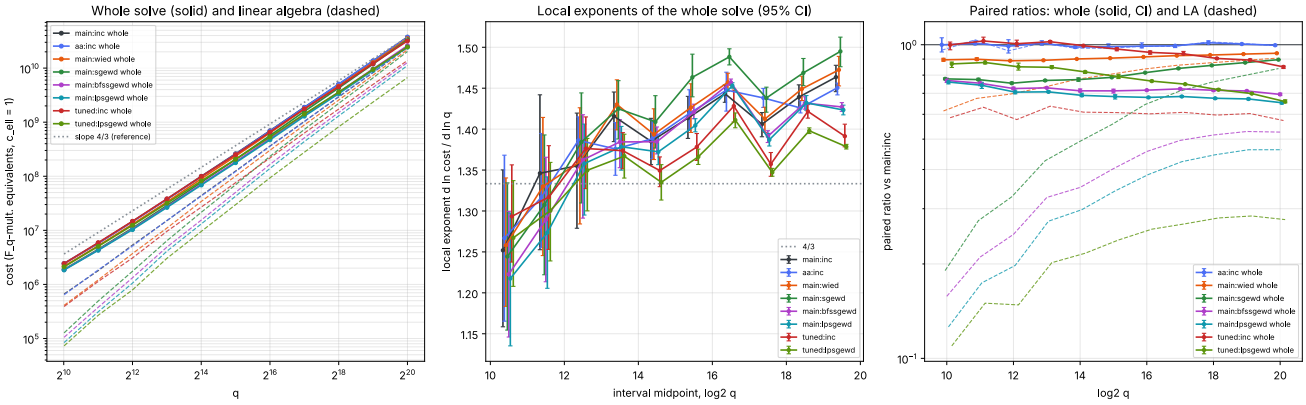
- Every logarithm of every arm at b = 10..20 was verified (`k phi(G) = phi(T)`) and equals the planted k: 14 arms, 438 solves, 0 failures.
- At b = 21 the gate caught a bug in the candidate. On curve 0, where ell = 2^61, the `lpsgewd` arms failed all 8 checks: the Berlekamp-Massey accumulator overflowed (amendment A5). It was fixed and rerun; the failed records are kept.

Controls.

- **Gate 0**: `main:inc` reproduces item 04's IC-2LP records exactly on 33/33 solves (k, cost and linear-algebra counters).
- **A/A** (`aa:inc`, the incumbent on an independent walk stream):
  - the exponent difference over 10..20 is +0.000 [-0.004, +0.004], which passes;
  - over 14..20 it is +0.005 [+0.003, +0.008]: the 3-curve bootstrap undercovers, so read differences of up to about 0.008 over 14..20 as noise;
  - per-size ratios lie within +-1.7% of 1 (the same data as item 04's A/A).
- **A/A of the candidate** (`aa:lpsgewd` against `main:inc`): Delta -0.020 [-0.024, -0.016] and ratio 0.655 at 2^20, which agrees with `main:lpsgewd`.

Rho reference (item 04's records on the same instances; not part of any verdict):

- At 2^20, measured negation-map rho costs 2.01e10 and the model costs 1.46e10.
- The best arm here costs 2.44e10, which is 1.21 times measured rho. At the 19-20 local exponents (1.38-1.42 against rho's 1.5), the crossover with measured rho would be near 2^22 to 2^23. That is an extrapolation, not a measurement.



cost, local exponents and paired ratios

Left: geometric-mean cost per size (3 curves) of the whole solve (solid) and of the linear algebra (dashed), in `F_q`-multiplication equivalents with `c_ell = 1`, with a slope-4/3 reference line. Middle: local exponents of the whole solve between consecutive sizes, with 95% bootstrap CIs. Right: paired ratios against `main:inc` per size, whole (solid, CI) and linear algebra (dashed). Run `results/run`, b = 10..20.

Per-size numbers (main arms)

Geometric means over curves 0-2; ratios are paired against `main:inc`, with 95% percentile bootstrap CIs over curves (2000 resamples).

log2 q	`inc` cost	`inc` LA share	`inc` solver cols, nnz/col	`lpsgewd` cost	ratio whole	ratio LA	`lpsgewd` solver cols, nnz/col	`tuned:lpsgewd` ratio	`tuned:inc` ratio	A/A ratio
10	2.43e6	0.27	175, 7.3	1.85e6	0.760 [0.750, 0.769]	0.126	56, 10.4	0.869	0.999	0.999 [0.948, 1.052]
11	5.76e6	0.31	275, 8.4	4.28e6	0.743 [0.726, 0.759]	0.174	107, 11.6	0.878	1.028	1.009 [1.000, 1.029]
12	1.46e7	0.37	456, 9.4	1.03e7	0.706 [0.697, 0.717]	0.198	195, 12.3	0.850	1.007	0.989 [0.941, 1.038]
13	3.75e7	0.40	740, 10.3	2.65e7	0.708 [0.704, 0.713]	0.274	373, 13.4	0.847	1.022	1.011 [0.997, 1.030]
14	9.99e7	0.44	1190, 12.1	6.89e7	0.690 [0.683, 0.700]	0.298	656, 14.1	0.819	0.993	0.983 [0.975, 0.989]
15	2.60e8	0.48	1924, 13.4	1.78e8	0.683 [0.671, 0.693]	0.342	1169, 14.5	0.791	0.969	0.985 [0.973, 1.006]

log2 q	inc cost	inc LA share	inc solver cols, nnz/col	lpsgewd cost	ratio whole	ratio LA	lpsgewd solver cols, nnz/col	tuned:lpsgewd ratio	tuned:inc ratio	A/A ratio
16	6.93e8	0.52	3114, 15.2	4.71e8	0.679 [0.676, 0.681]	0.386	2082, 15.0	0.766	0.945	0.989 [0.977, 1.014]
17	1.89e9	0.55	5022, 17.0	1.29e9	0.684 [0.681, 0.688]	0.424	3671, 15.2	0.749	0.935	0.992 [0.979, 0.999]
18	5.00e9	0.59	8029, 19.2	3.37e9	0.675 [0.670, 0.678]	0.446	6322, 15.3	0.719	0.904	1.015 [1.001, 1.030]
19	1.36e10	0.61	12835, 21.6	9.10e9	0.672 [0.667, 0.675]	0.463	10843, 15.3	0.699	0.893	1.005 [1.001, 1.011]
20	3.74e10	0.65	20462, 25.3	2.44e10	0.653 [0.647, 0.660]	0.463	18601, 15.1	0.659	0.850	0.996 [0.992, 1.002]

Exponents (b = 10..20; every solve is a point; paired bootstrap over curves)

arm	whole 10-20	Delta vs inc	whole 14-20	Delta vs inc	LA 10-20	Delta vs inc	LA 14-20	whole c_ell = 9	Delta vs inc	relation phase
main:inc (incumbent)	1.398 [1.394, 1.402]	-	1.425 [1.422, 1.428]	-	1.519 [1.512, 1.525]	-	1.514	1.491 [1.485, 1.497]	-	1.295
aa:inc (A/A)	1.398 [1.394, 1.403]	+0.000 [-0.004, +0.004]	1.431	+0.005 [+0.003, +0.008]	1.521	+0.003 [-0.004, +0.010]	1.523	1.494	+0.002 [-0.004, +0.008]	1.294
main:lpsgewd	**1.380 [1.376, 1.384]**	**-0.018 [-0.020, -0.016]**	**1.415 [1.413, 1.417]**	-0.011 [-0.013, -0.008]	1.702 [1.697, 1.707]	+0.184 [+0.179, +0.189]	1.621	1.563	+0.071 [+0.068, +0.075]	1.295
aa:lpsgewd	1.378 [1.375, 1.382]	-0.020 [-0.024, -0.016]	1.418	-0.007 [-0.010, -0.005]	1.694	+0.176	1.624	1.559	+0.068	1.294
main:bfssgewd	1.388 [1.384, 1.391]	-0.010 [-0.012, -0.009]	1.421	-0.004 [-0.007, -0.002]	1.689	+0.171	1.612	1.568	+0.077	1.295
main:lpsgelz	1.389 [1.385, 1.393]	-0.009 [-0.011, -0.007]	1.424	-0.002 [-0.004, +0.001]	1.700	+0.181	1.619	1.573	+0.082	1.295
main:wied	1.406 [1.402, 1.410]	+0.008 [+0.007, +0.009]	1.435	+0.010 [+0.010, +0.011]	1.573	+0.054	1.551	1.530	+0.038	1.295
main:sgewd	1.422 [1.418, 1.426]	+0.024 [+0.023, +0.025]	1.462	+0.037	1.722	+0.203	1.644	1.615	+0.123	1.295
tuned:inc (kappa 1.41)	1.373 [1.370, 1.376]	-0.025 [-0.028, -0.022]	1.391	-0.035 [-0.037, -0.033]	1.515	-0.004 [-0.010, +0.003]	1.504	1.470	-0.021 [-0.026, -0.016]	1.315
tuned:lpsgewd (kappa 1.41)	**1.357 [1.354, 1.360]**	**-0.041 [-0.043, -0.039]**	**1.375 [1.373, 1.378]**	-0.050 [-0.052, -0.047]	1.648	+0.129	1.575	1.493	+0.002 [-0.002, +0.005]	1.315

The arms run to 2^18 only (bfslz, bfswd, sgelz, bfssgelz; their Deltas are over 10..18 against inc over 10..18) are in analysis.md.

Local exponents of the whole solve (selected; all arms in analysis.md):

interval	inc	lpsgewd	bfssgewd	tuned:inc	tuned:lpsgewd
14-15	1.385 [1.357, 1.413]	1.372 [1.357, 1.392]	1.385	1.349	1.335
15-16	1.413 [1.389, 1.440]	1.405 [1.397, 1.413]	1.421	1.378	1.366
16-17	1.443 [1.433, 1.454]	1.453 [1.450, 1.457]	1.457	1.428	1.411
17-18	1.406 [1.391, 1.420]	1.387 [1.379, 1.395]	1.391	1.357	1.347
18-19	1.439 [1.426, 1.454]	1.432 [1.423, 1.441]	1.432	1.422 [1.413, 1.430]	1.398 [1.395, 1.402]
19-20	1.464 [1.446, 1.478]	1.423 [1.417, 1.429]	1.427	1.391 [1.378, 1.406]	1.379 [1.376, 1.381]
trend per bit	+0.018 [+0.014, +0.022]	+0.020 [+0.017, +0.025]	+0.020 [+0.017, +0.023]	+0.010 [+0.008, +0.013]	+0.011 [+0.008, +0.013]

Linear-algebra local exponents at 19-20: inc 1.543 [1.515, 1.566], lpsgewd 1.543 [1.533, 1.553], tuned:lpsgewd 1.504 [1.499, 1.510]. Trend of the LA local exponents: inc +0.002 [-0.005, +0.007]; lpsgewd -0.041 [-0.047, -0.037], falling from SGE's large small-q advantage toward about 1.54.

Extension to 2^21 (amendments A4, A5; not in the primary fits)

The extension covers 3 curves for each of main:inc, main:lpsgewd, tuned:inc and tuned:lpsgewd; all 36/36 solves over 10..21 are verified. Tables: analysis_b21.md.

arm	cost at 2^21	ratio vs main:inc [95%]	LA ratio	solver cols, nnz/col	local exponent 20-21, whole	local exponent 20-21, LA	slope 14..21	Delta 14..21 vs inc
main:inc	1.003e11	-	-	32616, 28.0	1.425 [1.404, 1.447]	1.477 [1.444, 1.512]	1.428 [1.425, 1.430]	-
main:lpsgewd	6.54e10	0.652 [0.643, 0.659]	0.479 [0.470, 0.488]	31466, 15.2	1.421 [1.418, 1.424]	1.528 [1.519, 1.533]	1.416 [1.415, 1.418]	-0.012 [-0.014, -0.009]
tuned:inc	8.47e10	0.844 [0.835, 0.857]	0.588	23061, 33.6	1.415 [1.399, 1.429]	1.515	1.393 [1.392, 1.395]	-0.034 [-0.036, -0.032]

arm	cost at 2^21	ratio vs `main:inc` [95%]	LA ratio	solver cols, nnz/col	local exponent 20-21, whole	local exponent 20-21, LA	slope 14..21	Delta 14..21 vs inc
`tuned:lpsgewd`	**6.40e10**	**0.638 [0.631, 0.645]**	0.281	24021, 15.3	**1.379 [1.378, 1.381]**	1.496	**1.377 [1.375, 1.378]**	-0.051 [-0.053, -0.049]

- At 2^21, kappa 1.41 with `lpsgewd` is the cheapest arm. It overtakes `main:lpsgewd`, as the local exponents at 19-20 predicted.
- The incumbent's local exponent at 20-21 (1.425) is lower than at 19-20 (1.464). With 3 curves its CI is wide, so the incumbent's rise is not monotone point by point.
- `main:lpsgewd`'s LA ratio keeps creeping up, from 0.463 to 0.479.

Tuning (training curves 5-7, never in the fits)

- **Stage 1** (results/tune1/tune1.md; 10 arms x cx in {0, 2, 8}, b = 14, 16, 18, kappa = 2):
  - at b = 18 the geometric-mean whole cost is lowest for `lpsgewd`, cx 8: 3.371e9, against 4.995e9 for `inc`;
  - next come `lpsgelz` (3.571e9) and `bfssgewd` (3.574e9);
  - cx = 8, the edge of the grid, was frozen.
- **Stage 2** (results/tune2/tune2.md; b = 20):
  - `lpsgewd`: kappa 1: 2.831e10; 1.41: 2.469e10; **2: 2.446e10**; 2.83: 2.791e10. So kappa* = 2.
  - `inc`: kappa 1: 3.269e10; **1.41: 3.204e10**; 2: 3.695e10; 2.83: 4.961e10.
  - All 3 curves were complete for every kappa before the analysis.

What was built

- **Shared collection.** One walk serves every arm. At each of the incumbent's check points, every arm still running solves its own system.
  - The counters of that solve go to the arm.
  - An arm's cost is the transfer and relation-phase counters up to the check where it succeeded, plus all of its own linear-algebra counters.
  - So arms differ only in the linear algebra, and an arm's cost does not depend on which other arms share the walk.
  - Gate 0 checks that this rewiring leaves the incumbent's records unchanged.
- **BFS cycle basis.**
  - The consumed large-prime relations are logged.
  - At a check, a breadth-first spanning tree of the large-prime graph from the "1" vertex gives one row per non-tree edge.
  - The rows have the same count and span as the incumbent's DFS substitution rows.
- **Large primes as columns.** Every consumed relation becomes a row with its large primes as extra columns.
- **SGE**, with ideas ported from `koblitz_sparse_la.rs`:
  - singleton cascade;
  - Markowitz-order merging: a column is eliminated while its fill is below the average column weight + cx;
  - removal of the heaviest excess rows with a singleton cascade;
  - the k column is never eliminated, so no back-substitution is needed.
  - A debug check (`CHECK_SGE=1`) confirms that every reduced row is satisfied by the full Lanczos solution.
- **Wiedemann** on the square remainder, computing only the k coordinate:
  - Berlekamp-Massey on `s_j = (A^j b)[k]` with early termination;
  - `k = -(1/g_0) sum g_i s_{i-1}`;
  - so there is no second Krylov pass.
- **Counting.**
  - Every product mod ell is counted, in the incumbent's unit (one `M_ell` per stored nonzero per matrix-vector product, as Lanczos is charged).
  - Products by +-1 in SGE are additions and are not charged, as in the incumbent's own substitution, which is uncharged.
  - In practice SGE costs only tens of `M_ell`, because almost all its pivots and multipliers are +-1.
  - Inversions mod ell are not priced, as in item 04. `lpsgewd` makes fewer of them than `inc` (18.8k against 40.9k at 2^20).

Deviations and limits

- **Host load.** The host ran at load 8-14 on 4 cores, shared by three items (amendment A2).
  - The first b = 20 jobs, with all ten arms in one collection, would have passed the 60-minute cap, so they were stopped.
  - At b >= 19 the arms were split over two collections per curve. This leaves every arm's cost unchanged, and gate 0 confirms it for `inc`.
  - The four arms dominated in tuning (`sgelz`, `bfslz`, `bfswd`, `bfssgelz`) were not run at b >= 19.
  - As a result, `bfslz`, the only arm that lowered the LA exponent, has data only to 2^18.
- **Stage-2 grid changed** (amendment A1): kappa 0.5 and b = 16 were dropped and 2.83 was added. The b = 18 part was stopped.
- **The `tuned` stream (kappa 1.41) is exploratory** (amendment A3). It was not chosen by the frozen rule.
- **b = 21 is an extension** (A4) for `inc` and `lpsgewd` in both streams. It hit the BM overflow bug on curve 0, which was fixed and rerun (A5).
- **The A/A per-size CIs exclude 1 at 3 of 11 sizes**, by at most 1.7%, as in item 04.
  - A 3-curve percentile bootstrap undercovers.
  - All candidate effects at the constant level (0.65-0.94) are far outside this floor.
- **cx = 8 is the edge of its grid.** A larger cx may be slightly better.
- **The large-prime bound is not tuned.** As in item 04, every non-factor-base prime is a large prime.
- **The relation phase is unchanged.** Its exponent is 1.295 at kappa = 2 and 1.315 at kappa = 1.41.
  - It is now the larger share at every size for `lpsgewd` (54% at 2^20).
  - The remaining distance to 4/3 is therefore partly in the walk.
- Wall times come from a loaded shared host and are not part of any claim.

Reproduce

```
cargo build --release --example ic_improve2_01_genus3_linalg --example ic_improve2_01_analyze
cd research/ic_improve_20261015/01-genus3-linalg
A=../../../target/release/examples/ic_improve2_01_analyze
PAR=3 sh run_tune1.sh results/tune1 && $A tune1 results/tune1      # stage 1
PAR=2 sh run_tune2.sh results/tune2 && $A tune2 results/tune2      # stage 2
PAR=3 sh run_all.sh results/run                                    # main + A/A, b = 10..20
KAPPA=1.41 PAR=2 sh run_tuned.sh results/run                       # exploratory kappa 1.41
```

```
sh run_b21.sh results/run; for c in 0 1 2; do sh job.sh results/run - --bits 21 --curve $c --cx 8 \
  --kappa 1.41 --streams tuned --las inc,lpsgewd --out results/run/b21_c${c}_tuned.jsonl; done
$A gate results/run ../../ic_improve_20261014/04-genus3-2lp/results/run-20261014
$A analyze results/run 10 20 14 results/run/summary.json > results/run/analysis.md
$A analyze results/run 10 21 14 results/run/summary_b21.json > results/run/analysis_b21.md
python3 -I plot.py results/run main:inc aa:inc main:wied main:sgewd main:bfssgewd main:lpsgewd tuned:inc tuned:lpsgewd
```

Toolchain, host and binary hashes: results/run/env.txt.